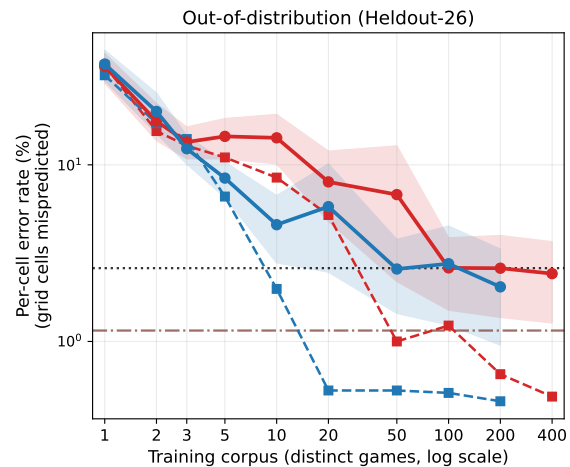
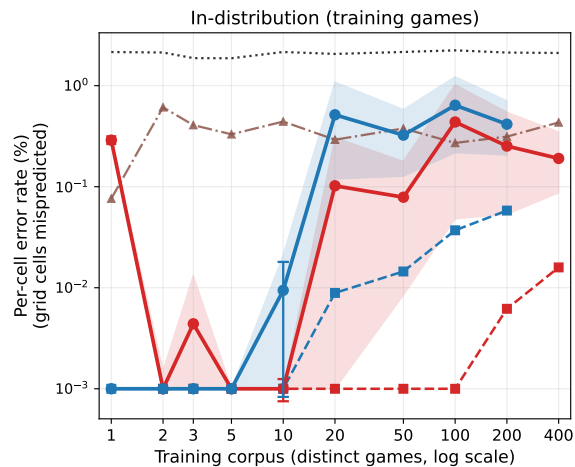




● Rule-conditioned (mean) ■ Rule-conditioned (median) ● Unconditional (mean) ■ Unconditional (median) ▲ no-rule (force-only movement) identity (copy last state)

Data-scaling of the NCA world model vs. training-corpus size. Each panel plots per-cell error rate (the fraction of grid cells whose predicted object-stack differs from the ground truth, averaged over teacher-forced rollout steps and over games) against the number of distinct training games, selected in ascending rule-count order from the token-deduplicated pool (the `n_per_rule` sweep), on log-log axes. *Left:* in-distribution, aggregated over each corpus’s own training games. *Right:* out-of-distribution, aggregated over the fixed Heldout-26 set of unseen games. Two parameter-matched models are compared: the rule-conditioned model ($n_h=256$ plus the text-encoder/decoder/slot stack, $\approx 15.97\text{M}$ parameters) and an unconditional model given slightly more capacity ($n_h=288$, $\approx 16.61\text{M}$), so any conditioning benefit cannot be attributed to parameter count. Solid lines are the mean over games (or heldout pairs); dashed lines are the median. The shaded band is a bootstrap 95% confidence interval of the mean *across games* (left) or *across heldout pairs* (right). The capped vertical error bars show the *across-seed* spread (min-max of the per-seed mean), drawn only at corpus sizes with at least two training seeds; they measure run-to-run reproducibility and are distinct from the across-game band. Two model-independent references are shown. The dotted black line is the identity (copy-last-state) baseline—per-corpus state churn in-distribution ($\approx 2\%$) and 2.60% on Heldout-26. The brown dash-dot line is the no-rule baseline, which predicts the next state by stepping the game with its rules removed: any player object moves in the direction of the applied force unless a collision blocks it, and nothing else moves. This isolates plain force-driven movement from rule-driven behavior, and is a much stronger reference than identity: $\approx 0.4\%$ per corpus in-distribution and 1.15% on Heldout-26. In-distribution error stays near zero and dilutes only mildly as the corpus widens at fixed capacity, with the rule-conditioned mean degrading more slowly than the unconditional one. Out-of-distribution mean error falls toward the identity baseline as the corpus grows but stays above the no-rule baseline, so on average the model still trails a movement-only predictor on unseen games, even though its median drops below the no-rule baseline at the largest corpus.